In [2]:
```
#1.Sum of Even and Odd Numbers Get a number n from the user
# and calculate the sum of all even numbers and
# the sum of all odd numbers between 1 and n.
num=int(input("enter the number"))
es=0
ods=0
for i in range(1,num+1):
    if(i%2==0):
        es=es+i

    else:
        ods=ods+i
print(f"The even sum is:{es}\nThe odd sum is:{ods}")
```

```
The even sum is:110
The odd sum is:100
```

In [4]:
```
#2.Count Digits in a Number Get an integer from the user and count the number of
#Do not convert the number into a string. Example: 58392 → 5 digits
num=int(input("enter the numbers"))
count=0
for i in range(len(str(num))):
    count+=1
print(count,"digits")
```

```
6 digits
```

In [5]:
```
3.#Find the Second-Largest Number Given a list of distinct numbers, find the sec
l1=[1,22,3,41,5,64,7]
l=l1[0]
sl=l1[0]
for i in l1:
    if i>l:
        sl=l
        l=i
    elif i>sl and i!=l:
        sl=i
print("second largest number is:",sl)
```

```
second largest number is: 41
```

In [6]:
```
#4.Count Vowels and Consonants Get a string from the user and count the vowels a
#Ignore spaces, digits, and special characters.
n=input("enter the any string:")
v=0
c=0
for i in n:
    if i.isalpha():
        if i.lower() in "aeiou":
            v+=1
        else:
            c+=1
print("Number of vowels:",v)
print("Number od consonants:",c)
```

```
Number of vowels: 3
Number od consonants: 4
```

In [7]:
```
#5.Calculate Factorial Get a positive integer from the user and calculate its fa
num=int(input("enter the number:"))
```

```
fact=1
for i in range(num,0,-1):
    fact=fact*i
print("factorial:",fact)
```

```
factorial: 5040
```

In [8]:
```
#6.Find Numbers Divisible by 7 but Not by 5 Print all numbers between 1 and 200
#Also, count the matching numbers.
count=0
for i in range(1,201):
    if i%7==0 and i%5!=0:
        print(i)
        count+=1
print("Total count:",count)
```

```
7
14
21
28
42
49
56
63
77
84
91
98
112
119
126
133
147
154
161
168
182
189
196
Total count: 23
```

In [10]:
```
#7. Calculate Electricity Bill Get the units consumed by each of five customers
# * First 100 units: ₹2 per unit
# * Next 100 units: ₹3 per unit
# * Above 200 units: ₹5 per unit
units=[80,150,250,100,300]
for i in range(5):
    u=units[i]
    if u<=100:
        bill=u+2
    elif u<=200:
        bill=(100*2)+(u-100)*3
    else:
        bill=(100*2)+(100*3)+((u-200)*5)
    print("Customer",i+1,"Units:",u,"Bill:Rs",bill)
```

```
Customer 1 Units: 80 Bill:Rs 82
Customer 2 Units: 150 Bill:Rs 350
Customer 3 Units: 250 Bill:Rs 750
Customer 4 Units: 100 Bill:Rs 102
Customer 5 Units: 300 Bill:Rs 1000
```

In [12]:
```
#8.Reverse a Number Get an integer from the user and reverse its digits using a
#Do not convert it into a string. Example: 12345 → 54321
n=int(input("Enter a number:"))
reverse=0
for i in range(1):
    while n>0:
        digit=n%10
        reverse=reverse*10+digit
        n=n//10
print("reversed number:",reverse)
```

```
reversed number: 54321
```

In [14]:
```
#9.Analyze Student Marks Given a list of marks, count the students who passed (5
marks=[85,45,72,90,35,60,78]
passed=0
failed=0
distinction=0
for mark in marks:
    if mark>=50:
        passed+=1
    else:
        failed+=1
    if mark>=75:
        distinction+=1
print("number of student passed:",passed)
print("number of student failed:",failed)
print("number of student with distinction:",distinction)
```

```
number of student passed: 5
number of student failed: 2
number of student with distinction: 3
```

In [15]:
```
#10.Find Perfect Numbers Print all perfect numbers between 1 and 1000 using a si
#A perfect number equals the sum of its proper divisors. Example: 6 is perfect b
for num in range(1,1001):
    total=0
    for i in range(1,num):
        if num%i==0:
            total+=i
    if total==num:
        print(num)
```

```
6
28
496
```

In [ ]:
```

```